COURSE NAME			GRAPH THEORY	L	T	P	C
COURSE CODE			YMA103	4	1	0	5
C	P	A		L	T	P	H
5	0	0		4	1	0	5
PREREQUISITE			Basic concepts of Graphs				

On successful completion of this course, the students will be able to:

	COURSE OUTCOMES	DOMAIN	LEVEL
CO 1	**Explain** basic concepts of **graphs**	Cognitive	Understanding
CO 2	**Explain** vertex connectivity and edge connectivity in graphs	Cognitive	Understanding
CO 3	**Explain** Eulerian Graphs and Hamiltonian Graphs	Cognitive	Understanding
CO 4	**Apply** colouring principle for solving problems in Vertex colourings and Edge coloring	Cognitive	Applying
CO 5	**Demonstrate** planar graphs	Cognitive	Understanding

UNIT 1	**Basic Results**	12+3

Basic Concepts - Subgraphs - Degrees of Vertices - Paths and Connectedness Operations on Graphs - Directed Graphs: Basic Concepts - Tournaments.

UNIT 2	**Connectivity**	12+3

Vertex Cuts and Edge Cuts - Connectivity and Edge - Connectivity, Trees:Definitions, Characterization and Simple Properties - Counting the Number of Spanning Trees - Cayley's Formula.

UNIT 3	**Independent Sets and Matchings**	12+3

Vertex Independent Sets and Vertex Coverings - Edge Independent Sets -Matchings and Factors - Eulerian Graphs - Hamiltonian Graphs.

UNIT 4	**Graph Colourings**	12+3

Vertex Colouring - Critical Graphs - Triangle - Free Graphs - Edge Colourings of Graphs - Chromatic Polynomials.

UNIT 5	**Planarity**	12+3

Planar and Nonplanar Graphs - Euler Formula and its Consequences - K5 and K3,3 are Nonplanar Graphs - Dual of a Plane Graph - The Four-Colour Theorem and the Heawood Five-Colour Theorem- Kuratowski's Theorem.

LECTURE	60	TUTORIAL	15	PRACTICAL	0	TOTAL	75

TEXT BOOK
1.Douglas B. West, “Introduction to Graph Theory”, Prentice Hall of India, Second Edition, 2002.
REFERENCES
1. Bondy J. A, and Murty U. S. R., “Graph Theory”, Springer, 2008. 2. Balakrishnan R. and Ranganathan K., “A textbook of Graph Theory”, Springer, 2012. 3. Graham R.L., Rothschild B.L and Spencer J.H., “Ramsey Theory”, Wiley Publishers, Second Edition, 1990. 4. Biggs N., “Algebraic Graph Theory”, Cambridge Tracts in Mathematics 67, Cambridge University Press, 1994. MX8003 Algebraic Theory of Semigroups.

COs VS POs

	PO 1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**
CO 1	3	3	2	1	3	1	0	1	1
CO 2	3	3	3	2	3	1	1	1	1
CO 3	3	3	2	1	3	1	0	1	1
CO 4	3	3	3	2	3	1	1	1	1
CO 5	3	3	2	1	3	1	0	1	1
TOTAL	**15**	**15**	**12**	**7**	**15**	**5**	**2**	**5**	**5**
SCALED VALUE	**3**	**3**	**3**	**2**	**3**	**1**	**1**	**1**	**1**
0 - No Relation, 1 – Low Relation, 2- Medium Relation, 3- High Relation									
1-5→1, 6-10→2, 11-15→3									